MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, TRIAXIAL, 50 OHMS, M17/131-RG403

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

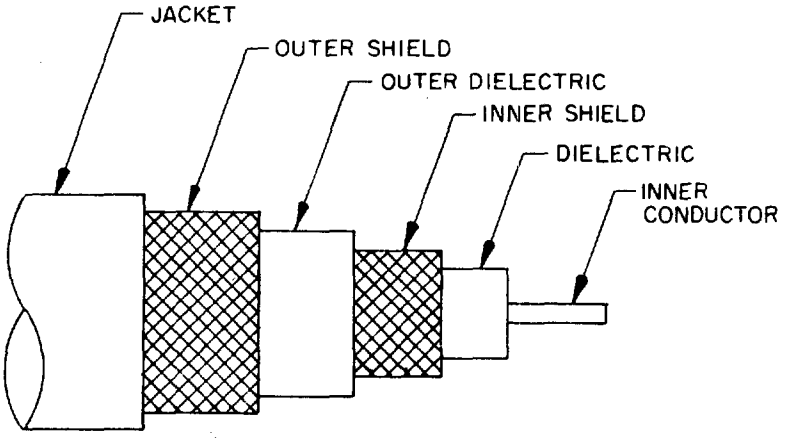


FIGURE 1. Configuration.

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TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of AWG size 38 silver-coated, copper-covered steel. Overall diameter: .012 inch nominal.
Dielectric	Type F-1. Overall diameter: .033 ±.002 inch.
Inner shield	Silver-coated, AWG size 38 copper wire. Overall diameter: .048 - .054 inch. Coverage: 95.9% nominal Carriers: 16 Ends: 3 Picks/inch: 25.0 ±10%
Outer dielectric	Type M-1. Overall diameter: .068 - .080 inch.
Outer shield	Silver-coated, AWG size 38 copper wire. Overall diameter: .084 - .098. Coverage: 95.9% nominal Carrier: 24 Ends: 7 Picks/inch: 15.4 ±10%
Outer jacket	Type IX. Overall diameter: .104 - .128 inch. Wall thickness: .010 inch minimum.

Engineering information:

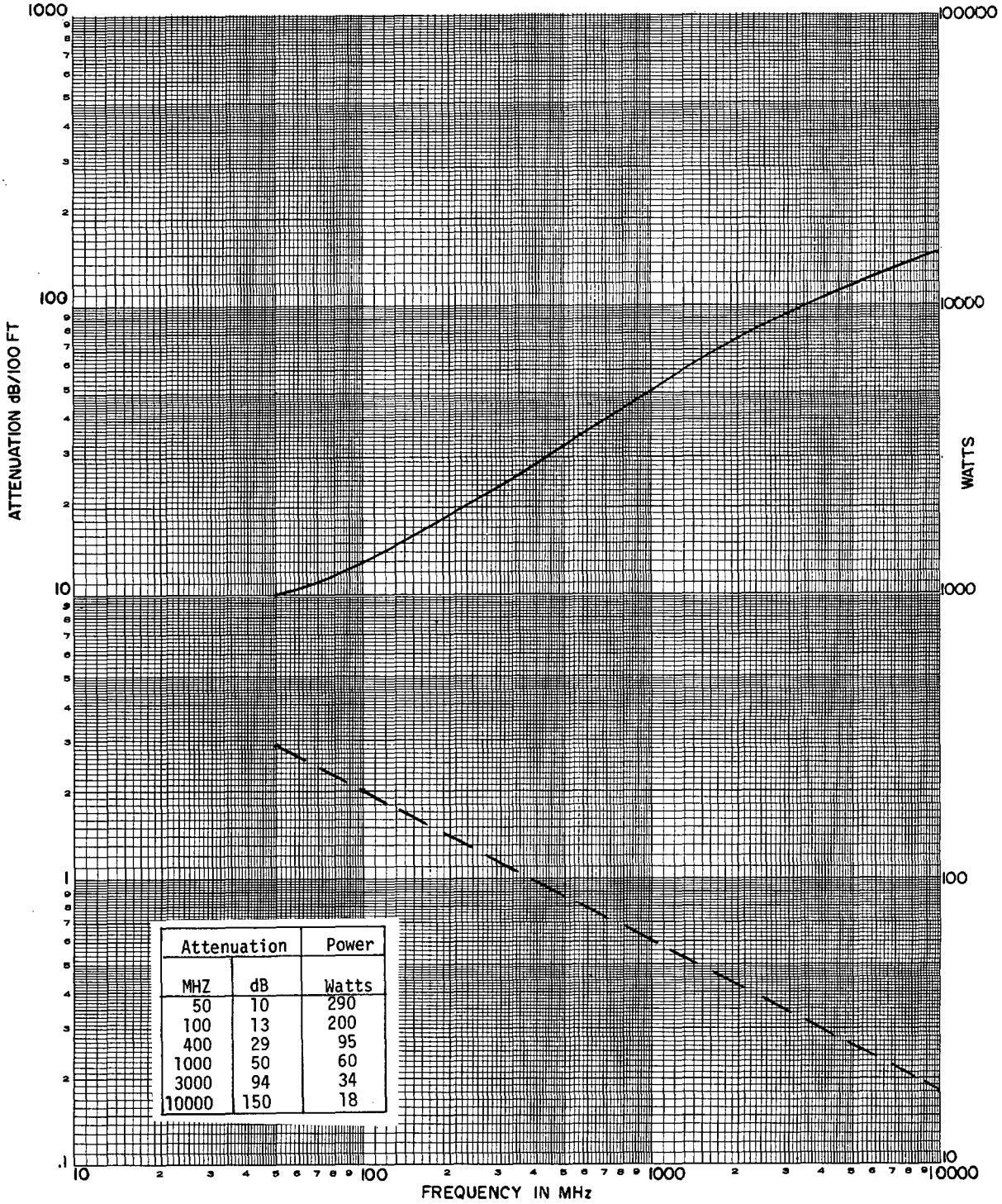
Continuous working voltage: 750 Vrms, minimum inner conductor to inner shield, 300 Vrms, minimum inner shield to outer shield.
Operating frequency: 10 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 24.45 ohms per 100 feet.
Elongation: 10 percent, minimum.
Tensile strength: 50 $klb_f/inch^2$, minimum.
Engineering notes: This cable useful in high temperature applications.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors:
Inner conductor to core: 1.5 pounds minimum, 4 pounds maximum.
Aging stability: Not applicable.
Stress crack resistance: +230° ±5°C for 96 hours.
Outer conductor integrity: Not applicable.
Cold bend: -55°C ±2°C.
Dimensional stability: +200° ±5°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Weight: 1.5 pounds per 100 feet, maximum.

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Attenuation		Power
MHZ	dB	Watts
50	10	290
100	13	200
400	29	95
1000	50	60
3000	94	34
10000	150	18

MAXIMUM ATTENUATION ——(Test requirements shall be as noted as line indicated on graph)
MAXIMUM POWER – – –
AT 25°C SEA LEVEL

FIGURE 2. Power rating and attenuation.

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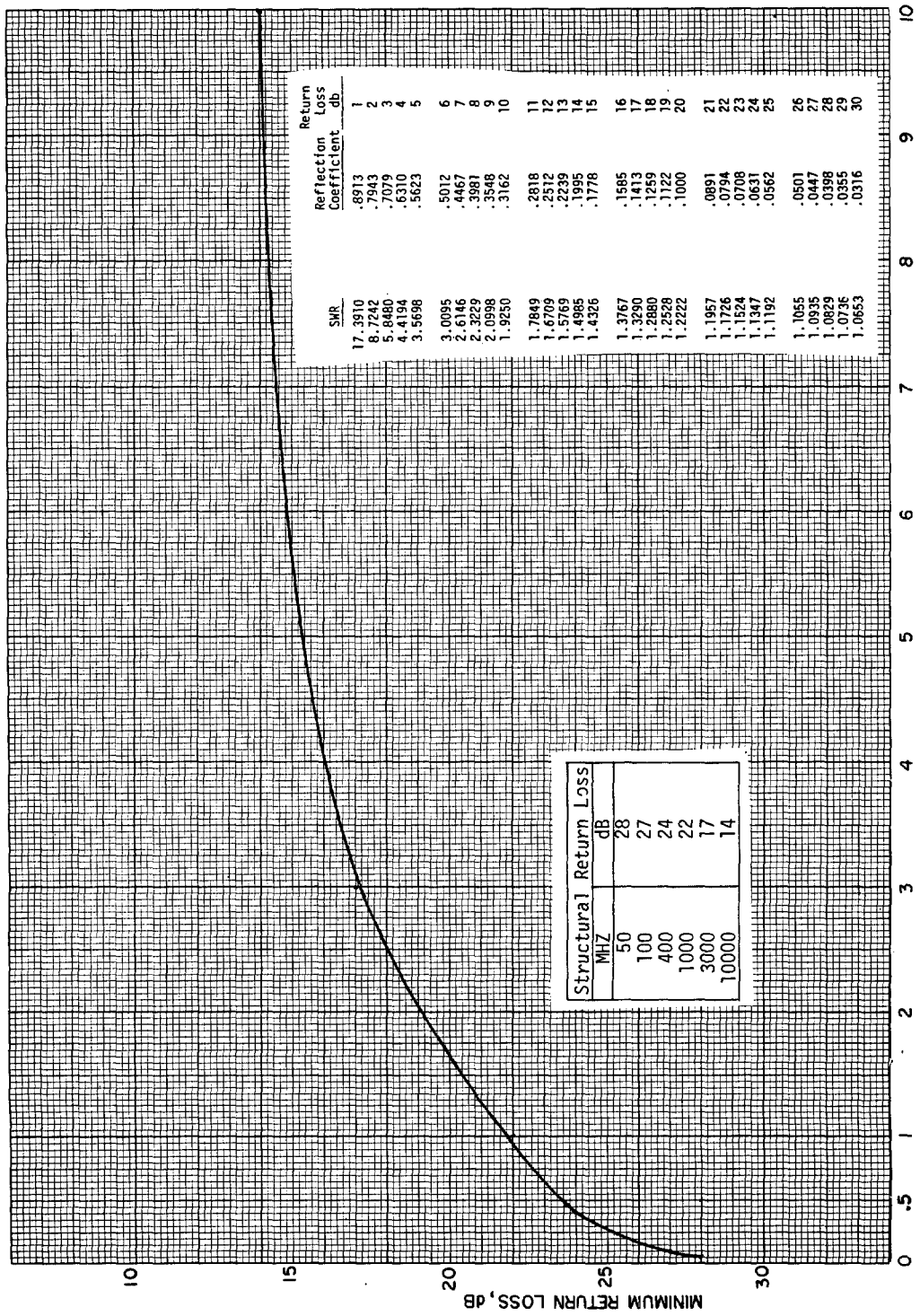


FIGURE 3. Structural return loss.

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Electrical:
Test frequency: 50 MHz to 10 GHz.
Spark test: 2,000 Vrms +25%, -0%.
Voltage withstanding: 2,000 Vrms inner conductor to inner shield.
1,000 Vrms inner shield to outer shield.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,000 Vrms minimum, inner conductor to inner shield.
Characteristic impedance: 50 ohms ±2.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 30.2 pF/foot, inner conductor to inner shield; 160 pF/foot, maximum inner shield to outer shield.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/131-RG403	RG-403/U

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS
User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-0716-21)

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MIL-C-17/131A

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